# Probabilistic Symptom-Pattern Screening over Paraphrased DSM-5 Criteria with a Neural Ensemble and an LLM Interface

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**Abstract.** We present a research prototype that estimates how closely a person's reported symptoms resemble twelve DSM-5-inspired conditions. A five-member multilayer-perceptron ensemble, trained on synthetic cases generated from paraphrased criteria, outputs calibrated multi-label probabilities; a large language model converts free text into structured symptoms and writes a grounded explanation. On held-out synthetic data the ensemble reaches a macro AUROC of 0.979 (95% CI 0.977–0.980), 0.004 above logistic regression, and degrades gracefully when half of the symptoms are withheld (0.910). Because training and test data come from the same generator, these figures measure internal consistency, not clinical validity; we state the limitations and safeguards explicitly.

## 1. Introduction

Structured diagnostic manuals such as the DSM-5 describe conditions as combinations of symptoms, durations and functional impairment. In practice, people describe their experience in free text, symptoms overlap across conditions, and comorbidity is common, so a single "best" label is often the wrong abstraction.

We ask a narrow, engineering question: can a small neural network, fed with a structured symptom report, return calibrated *probabilities* for several co-occurring conditions, and can a language model make that interaction natural without being trusted to produce the numbers? The system is deliberately a screening-style research prototype, not a diagnostic device.

We make four contributions:

1. A paraphrased, machine-readable knowledge base of 68 symptoms and 12 conditions, with an explicit core / common / occasional probability for each symptom-condition pair.
2. A synthetic case generator with comorbidity, severity, reporting noise, sub-threshold hard negatives, duration and impairment context.

3. A five-member MLP ensemble with temperature scaling and occlusion-based explanations, compared against logistic regression, with paired-bootstrap confidence intervals and a symptom-dropout robustness test.
4. An end-to-end system: a Persian right-to-left web interface, an LLM layer for symptom extraction and explanation, and a standalone offline build.

## 2. Related Work

**Symptom checklists.** Instruments such as the PHQ-9 and GAD-7 turn self-report into a severity score for one condition [1, 2]. They are validated, but each targets a single disorder and none returns a joint distribution over comorbid conditions. Field trials of the DSM-5 found that diagnostic reliability varied widely between conditions, and was low for major depressive disorder and generalized anxiety disorder [3]. This motivates reporting uncertainty instead of a single label.

**Calibration and ensembles.** Modern neural networks tend to be over-confident, and temperature scaling is a simple post-hoc correction [4]. Deep ensembles improve both accuracy and uncertainty estimates [5]. We use both, and report the spread across ensemble members to the user.

**Explanations.** Occlusion removes an input and measures the change in the output; it is a model-agnostic idea from computer vision [6]. For binary symptom inputs it is exact, cheap and easy for a non-expert to read, so we use it instead of gradient-based attributions.

**Language models in mental-health tools.** Language models are increasingly proposed for screening and support, yet they can produce fluent but unfounded clinical statements. We therefore restrict the model to two bounded roles: structured extraction under a closed vocabulary, and explanation of numbers it did not compute. All probability estimation stays in the network.

## 3. Data: Knowledge Base and Synthetic Generator

**Knowledge base.** The DSM-5 text is copyrighted by the American Psychiatric Association [7], so nothing is copied from it. We wrote short original labels for 68 symptoms in ten groups (mood, sleep and energy, cognition, anxiety, trauma, obsessions, attention, eating, alcohol), and for each of 12 conditions we assigned every relevant symptom a probability class: core (0.85), common (0.50) or occasional (0.20), with a few hand-set values. These numbers are modelling assumptions, not published statistics. The conditions are major

depressive disorder, persistent depressive disorder, generalized anxiety disorder, panic disorder, social anxiety disorder, post-traumatic stress disorder, obsessive-compulsive disorder, ADHD, bipolar disorder (manic and hypomanic spectrum), insomnia disorder, binge-eating disorder and alcohol use disorder.

**Generator.** Each case is drawn as follows. The number of true conditions k is 0, 1, 2 or 3 with probabilities 0.10, 0.52, 0.28 and 0.10. The first condition is uniform; later ones are sampled with weight proportional to their mean affinity to those already chosen, using 18 boosted pairs such as depression with anxiety or anxiety with insomnia. A severity scale is drawn and applied to the prototype probabilities:

```
s \sim \mathrm{Beta}(2.5, 2.5), \qquad \sigma = 0.55 + 0.9\, s
```

A symptom j is then present under a noisy-OR over the chosen set Z, with a background rate b_j (0.03 by default, higher for common complaints such as fatigue):

```
P(x_j = 1 \mid Z, \sigma) = 1 - (1 - b_j) \prod_{d \in Z} \bigl(1 -
\min(0.98,\ \sigma\, \pi_{dj})\bigr)
```

Reporting noise is added afterwards: a present symptom is dropped with probability 0.05 and an absent one appears with probability 0.01. For k = 0, 60% of cases mimic one condition at a fixed scale of 0.3 and carry no label, which gives the model sub-threshold hard negatives; the rest are healthy.

**Context features.** Each case also has a duration bucket (under 2 weeks, 2 weeks to 6 months, over 6 months) drawn from a per-condition prior, and an impairment flag that is true with probability 0.95 for labelled cases, 0.30 for sub-threshold cases and 0.10 for healthy ones. The input vector therefore has 68 + 3 + 1 = 72 dimensions, and the target is a 12-dimensional multi-label vector.

**Splits.** We generated 20,000 training, 4,000 validation and 6,000 test cases with different seeds. Per-condition prevalence is 10% to 14%, the mean number of labels is 1.38, and 10.2% of training cases have no label.

## 4. Method

**Network.** Each ensemble member is a multilayer perceptron with layers 72 → 96 → 48 → 12, ReLU activations and dropout 0.2 after each hidden layer. Weights use He initialisation. The 12 outputs are independent sigmoids, so several conditions can be probable at once

and an all-low output means "no clear pattern". We implemented the forward pass, back-propagation and the Adam optimiser [8] directly in NumPy, because the sandbox used for this work could not install PyTorch.

**Training.** The loss is multi-label binary cross-entropy with a small L2 penalty (1e-4), where z are the logits and T a temperature:

```
\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{d=1}^{12} \Bigl[ y_{id} \log p_{id} + (1 - y_{id}) \log (1 - p_{id}) \Bigr], \qquad p_{id} = \sigma\!\left(z_{id} / T\right)
```

We use Adam with learning rate 2e-3, batch size 128, up to 80 epochs and early stopping on validation loss with patience 10. Dropout follows [9]. Five members are trained with different seeds; training all five takes about 25 to 50 seconds on a CPU.

**Calibration and ensembling.** After training, each member's temperature T is fitted on the validation set by minimising cross-entropy over T in [0.2, 5] [4]. The fitted values were close to 1 (1.005 to 1.041). The final probability is the mean of the five calibrated outputs, and the standard deviation across members is shown to the user as model disagreement [5].

**Explanations.** For each reported symptom j and condition d we compute an occlusion effect, the drop in probability when that symptom alone is switched off [6]:

```
\Delta_{jd} = p_d(x) - p_d\bigl(x \text{ with } x_j = 0\bigr)
```

The up to three symptoms with the largest positive effect are listed as the main supporting symptoms. The probability that no modelled condition applies is the product of (1 - p_d) over all conditions.

**LLM layer.** The language model has two jobs and never produces a probability. (1) Extraction: the person's text, treated purely as data, is mapped to ids from the closed symptom catalogue, plus a duration bucket and an impairment flag, as strict JSON; any id outside the catalogue is discarded. (2) Explanation: given only the model's numbers and the reported symptoms, it writes a short Persian paragraph that must not add facts, give treatment advice or say that the person "has" a disorder. When no API key is configured, a deterministic keyword extractor and a template explanation take over, so the application still runs offline. Selecting the symptom "thoughts of death or self-harm" triggers a prominent safety message with emergency guidance.

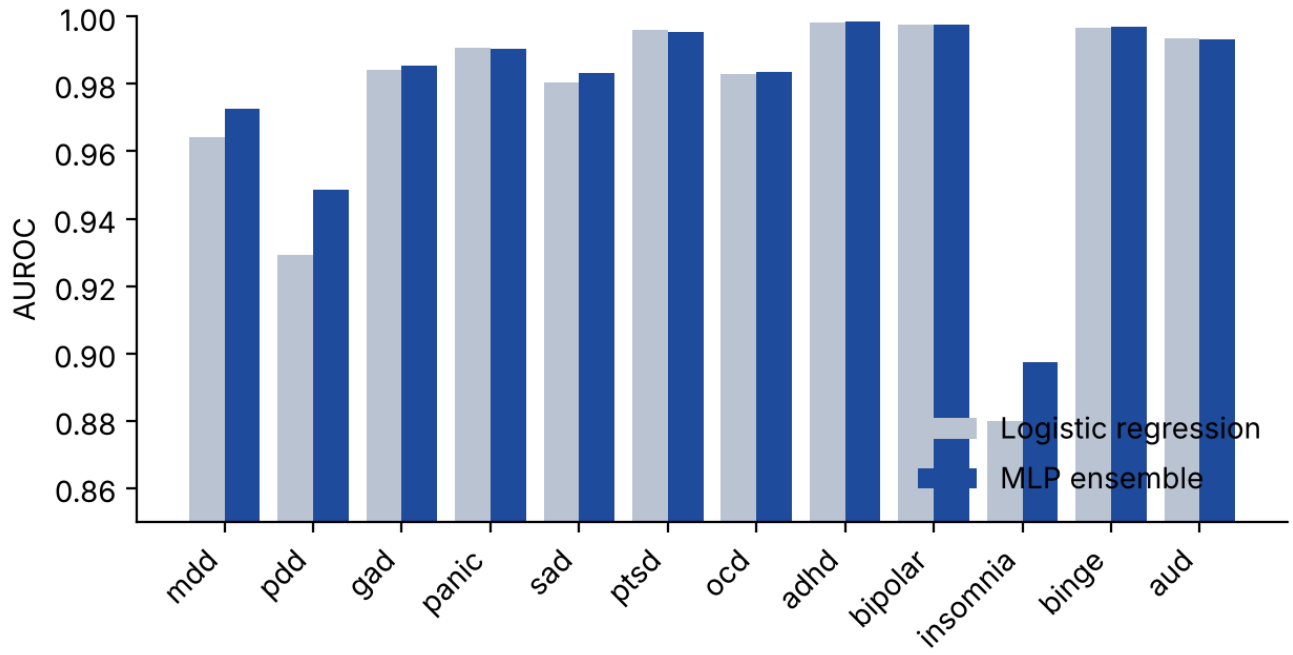
# 5. Experiments and Results

**Setup.** All numbers are on the 6,000 held-out synthetic test cases. The baseline is one-vs-rest logistic regression (C = 1) on the same 72 inputs. Macro metrics average over the 12 conditions; F1 uses a 0.5 threshold. Top-1 and top-3 hit rates are the share of labelled cases where the highest-ranked (or any of the three highest-ranked) condition is a true label. Confidence intervals come from 300 paired bootstrap resamples of the test cases.

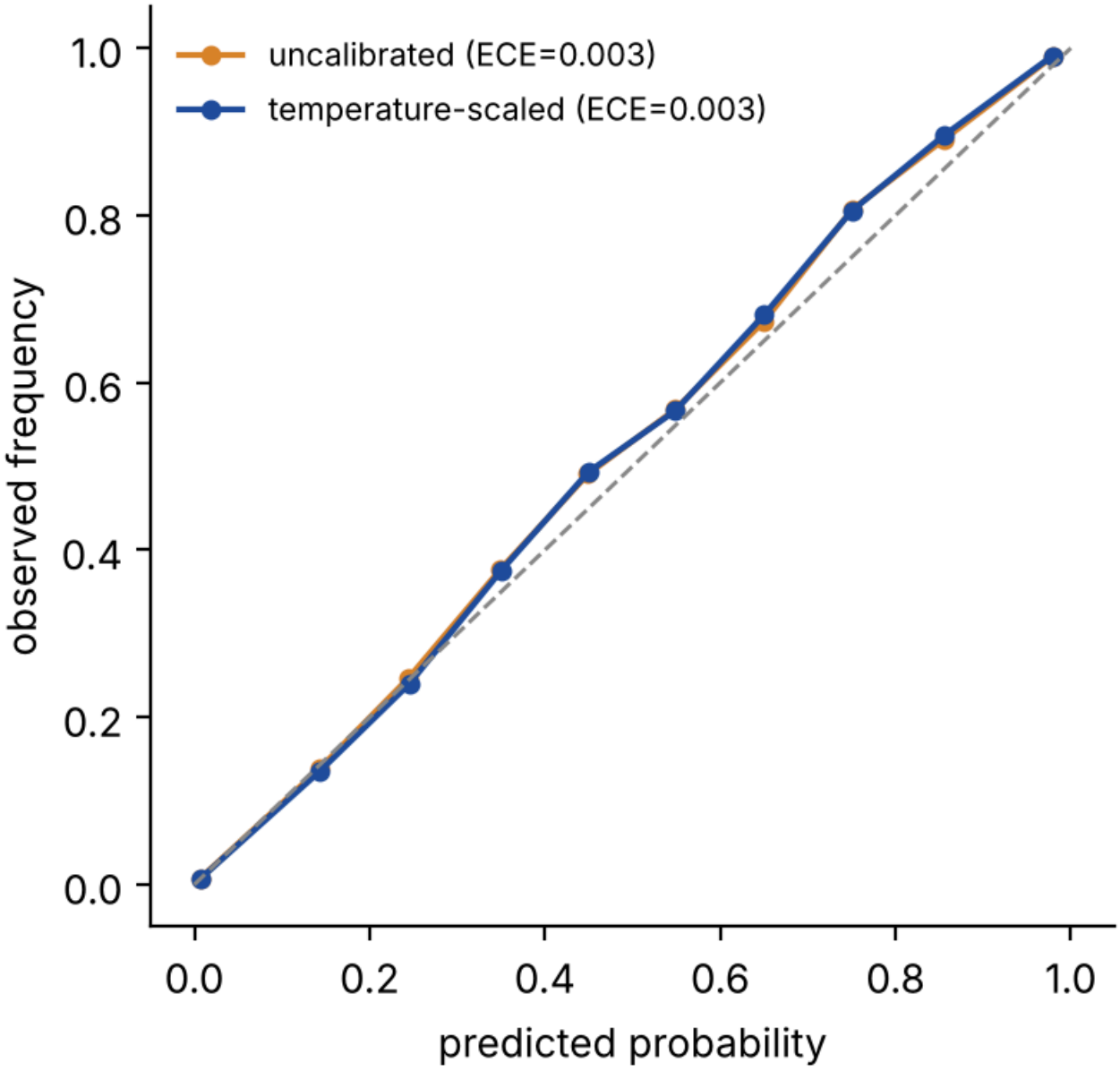
| Model | Macro AUROC | Macro AUPRC | Macro F1 | Top-1 hit | Top-3 hit | ECE |
|---|---|---|---|---|---|---|
| MLP ensemble (5), calibrated | 0.9786 | 0.9116 | 0.8441 | 0.9361 | 0.9915 | 0.0031 |
| MLP ensemble (5), uncalibrated | 0.9786 | 0.9116 | 0.8442 | 0.9361 | 0.9915 | 0.0032 |
| Single MLP, calibrated | 0.9773 | 0.9073 | 0.8429 | 0.9328 | 0.9909 | 0.0035 |
| Logistic regression | 0.9745 | 0.8980 | 0.8352 | 0.9329 | 0.9895 | 0.0021 |

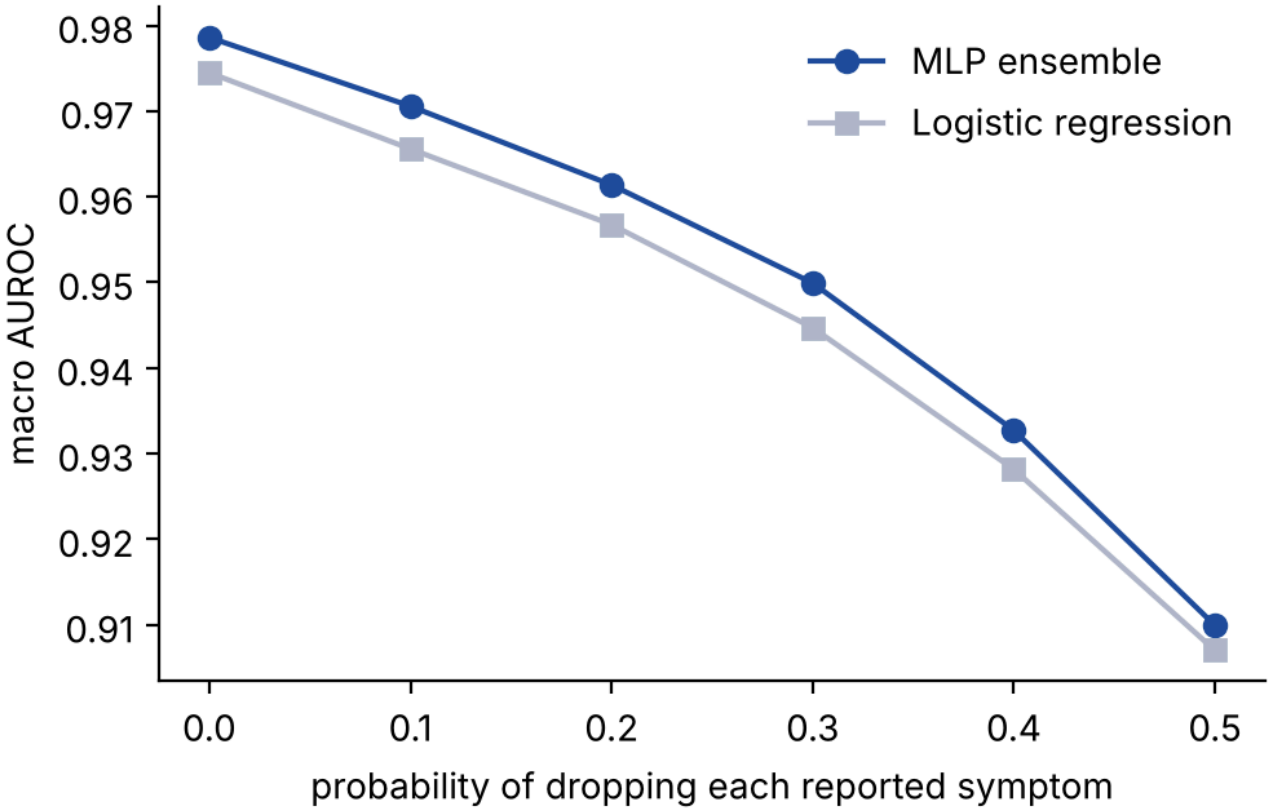
**Accuracy.** The ensemble beats logistic regression by 0.0041 macro AUROC, with a paired-bootstrap 95% CI of 0.0032 to 0.0051. The gain is small but consistently positive. A small gap is expected: the generator is a noisy-OR, which is close to log-linear, so there is little non-linear structure for a deeper model to exploit. The ensemble adds a further 0.0013 over a single network.

**Calibration.** All models are already well calibrated on this data. Temperature scaling moved ECE only from 0.0032 to 0.0031, and logistic regression has the lowest ECE (0.0021). We therefore do not claim that calibration is a benefit here; on real data, where the training distribution is less well specified, it is likely to matter more.

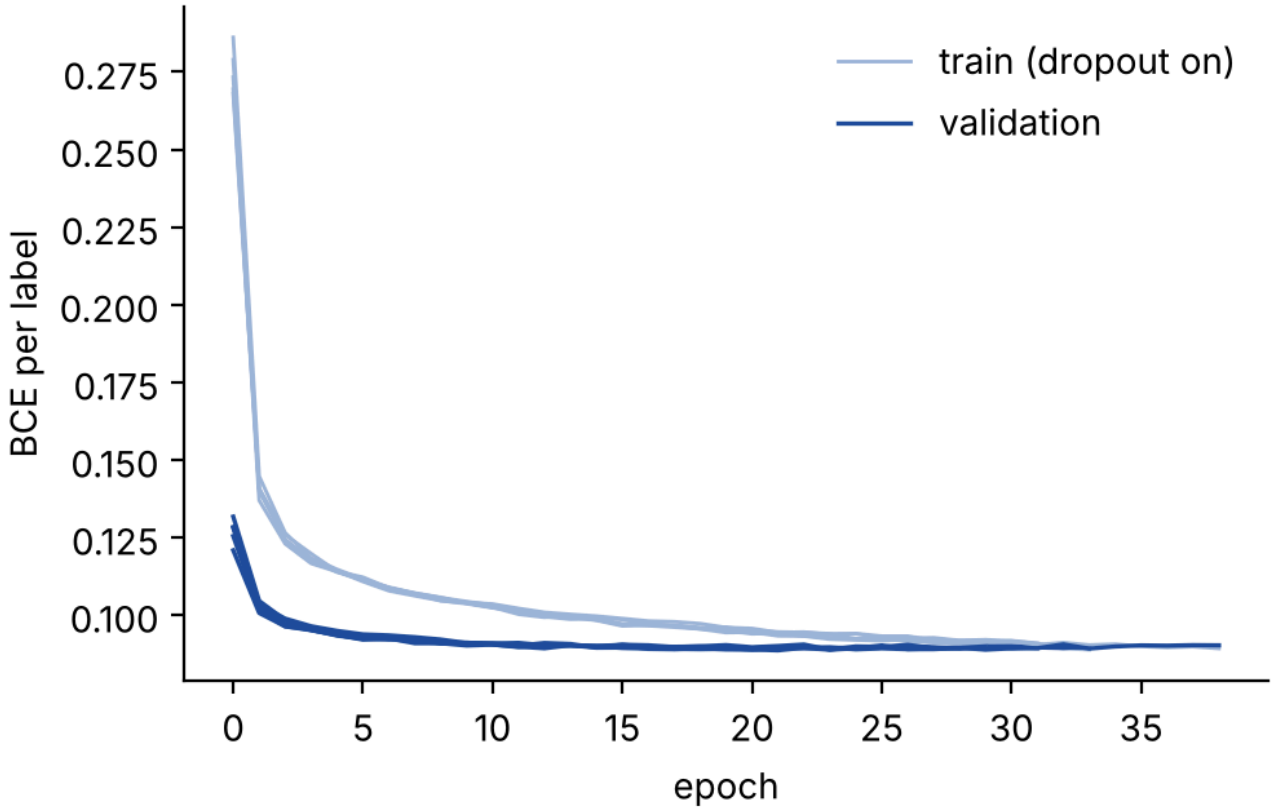



**Per condition.** Ten of twelve conditions reach AUROC of 0.97 or higher. The two hardest are insomnia disorder (0.897) and persistent depressive disorder (0.949), whose symptoms overlap heavily with those of other conditions; these are also where the network gains most over logistic regression (+0.018 and +0.019).


uncalibrated (ECE=0.003)
temperature-scaled (ECE=0.003)
observed frequency
predicted probability
0.0
0.2
0.4
0.6
0.8
1.0




**Robustness.** People under-report. When each reported symptom is dropped at random, macro AUROC falls smoothly from 0.979 (no dropout) to 0.950 at 30% and 0.910 at 50%, against 0.907 at 50% for logistic regression. Neither model collapses, and the ensemble keeps a small lead throughout.




**Training.** Validation loss stabilised after roughly 20 epochs, and early stopping ended training at 30 to 39 epochs per member.

## 6. System and User Interface

The request flow has five steps:

1. The person optionally writes a free-text description in Persian or English.
2. The LLM layer (or the keyword fallback) maps it to catalogue symptoms, a duration bucket and an impairment flag, which the person reviews and edits.
3. The ensemble computes probabilities, model disagreement, occlusion effects and the probability of no clear pattern.
4. The interface draws ranked bars with a strength label (strong, plausible, weak), the main supporting symptoms and the disagreement band.
5. The LLM layer, or a template, writes a short explanation grounded only in those numbers.

The interface is a single right-to-left Persian page with a calm, low-contrast palette, light and dark themes, grouped symptom chips and a persistent notice that the tool is not a diagnosis. The ensemble also runs in the browser in plain JavaScript, so results update instantly as symptoms are toggled. On a test case, the JavaScript and Python implementations returned identical probabilities and supporting symptoms. A server-free build embeds the exported weights (about 470 KB) in one HTML file; only free-text extraction and the LLM explanation need the Flask server.

## 7. Limitations and Ethics

**The evaluation is circular.** Training and test data come from the same generator built from the same knowledge base. The high scores show that the network recovers the assumptions we encoded; they say nothing about whether those assumptions match real patients. No clinical validation has been done.

**Probabilities are overconfident in practice.** A handful of core symptoms produces probabilities near 0.99 because, in the synthetic world, such combinations are almost unique to one condition. Real presentations are noisier, so the displayed numbers should be read as pattern similarity, not as the chance that a person has a condition.

**Scope.** Twelve conditions are modelled and many common ones are absent (for example personality, psychotic and neurodevelopmental disorders), so "no clear pattern" does not mean "nothing". Duration and impairment are coarse, symptoms are binary, and nothing models medical or substance causes of symptoms, which the DSM-5 requires clinicians to rule out.

**Copyright and provenance.** Labels are original paraphrases and the probabilities are our own assumptions. Any validation against official criteria should use a licensed copy of the manual.

**Safety and misuse.** The interface never presents a diagnosis and always points to professional care. Reporting thoughts of death or self-harm triggers an emergency message. The LLM is confined to extraction and explanation, and its output is validated against the closed catalogue. The tool must not be used for triage, insurance, employment, legal or school decisions, or to label other people.

**Cultural validity.** Symptom wording, help-seeking and diagnostic thresholds vary across cultures and languages. The Persian labels have not been tested with Persian-speaking clinicians or the public.

## 8. Future Work and Reproducibility

**Next steps.** The first priority is real data: clinician-rated cases from licensed structured interviews, used to refit the network and to measure calibration honestly. Further work should compare the output with PHQ-9 and GAD-7 scores, evaluate LLM extraction against clinician-annotated Persian text, add conformal prediction sets in place of single probabilities, model severity and course over time, and run a usability study. A PyTorch port would make the model easier to extend.

**Known gap.** In the environment where this prototype was built there was no API key, so the live LLM path was exercised only through its fallback; the keyword extractor, template explanation, API contract and safety flag are covered by tests.

**Reproduce.** The code needs Python 3 with numpy, scipy, scikit-learn, matplotlib and Flask. All seeds are fixed, so the numbers above are reproduced exactly.

```
python data/generate.py          # synthetic train/val/test sets
python model/train.py            # ensemble, calibration, metrics, figures,
model.json
python tests/test_pipeline.py    # vignettes, API, safety flag, extractor
export ANTHROPIC_API_KEY=...     # optional: enables the LLM layer
python backend/app.py            # UI at http://127.0.0.1:8000
python ui/build_standalone.py    # one-file offline build in dist/
```

## References

Reference details were written from memory and should be checked against the originals before submission.

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